🌌 Section: Our Dynamic Universe

🔷 Topic: The Expanding Universe

Question 5 – Redshift Demonstration

(a) Determine the wavelength represented by the students

✔3 marks

Speed of each student = $v = 1.5\,\mathrm{m/s}$
Time between wavefronts = $T = 3.5\,\mathrm{s}$

$\lambda = v \cdot T = 1.5 \times 3.5 = \ 5.25\,\mathrm{m}$

(b) Is the star moving towards or away from the observer?

✔2 marks

✅ **Answer:**

> The star is moving **away** from the observer.
> The observed wavefronts are spaced **further apart**, meaning **wavelength increases** (i.e. redshift).

🧠 Tip: More spaced = lower frequency = star is receding.

(c)(i) Explain how Olbers' Paradox supports the expanding Universe

✔2 marks

✅ **Answer:**

> If the Universe were **static and infinite**, the night sky would be **bright**, not dark.
> But the sky is dark — this supports the idea that the Universe is **expanding**, and light from distant stars **hasn't reached us yet** or has **redshifted beyond visible light**.

(c)(ii) State one other piece of evidence for an expanding Universe

✔1 mark

✅ **Answer:**

> The **Cosmic Microwave Background Radiation (CMBR)**

Also acceptable: Hubble's Law or abundance of light elements.

🧠 Revision Tips

- **Wavelength = speed × time** — even when the "wavefronts" are people!
- **Redshift** means longer observed wavelength: object is moving **away**
- **Olbers' Paradox**: a dark night sky contradicts a static, infinite Universe
- **CMBR** is radiation leftover from the Big Bang — a key clue for expansion

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